

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B. Tech

Branch: Civil Engineering

Semester: IV

Subject Code & Name: BTCVC405 & Hydraulics - II

Max Marks: 60

Date: 31/12/2024

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

	(Level /CO)	Marks
Q. 1 Objective type questions. (Compulsory Question)		12
1 The flow characteristics of a channel does not change with time at any point. What type of flow is it? a. Steady flow b. Uniform flow c. Laminar flow d. Turbulent flow	L1	1
2 A Cipolletti weir is particular type of trapezoidal weir, the sloping sides of which have an inclination of a. 1 horizontal to 4 vertical b. 1 horizontal to 5 vertical c. 1 horizontal to 3 vertical d. 1 horizontal to 2 vertical	L1	1
3 What is the dimension of C in Chezy's equation? a. LT b. $L^{1/2} T^{-1}$ c. LT^{-1} d. $L^{-1} T^{-1}$	L1	1
4 Calculate the specific energy for a rectangular channel having depth 2m and width 3m. Given: Discharge (Q) = $8.78 \text{ m}^3/\text{s}$. a. 2.11m b. 3.11m c. 4.11m d. 5.11m	L1	1
5 Which of the following assumptions about a GVF is false? a. Channel is prismatic b. Pressure distribution is hydrostatic c. Flow characteristics change with time d. Roughness coefficient is constant	L1	1
6 During a steady jump, the value of Froude lies in between _____ a. 1 to 2.5 b. 2.5 to 4.5 c. Less than 1 d. 4.5 to 9	L1	1
7 In a stationary vertical plate, the jet after striking the plate will move _____ a. In opposite direction b. Along the plate c. Perpendicular to the plate d. None of the above	L1	1

8	Which type of turbine is a Francis Turbine?	L1	1
	a. Impulse Turbine		
	b. Screw Turbine		
	c. Reaction turbine		
	d. Turgo turbine		
9	Specific speed of reaction turbine is between?	L1	1
	a. 5 and 50		
	b. 10 and 100		
	c. 100 and 150		
	d. 150 and 300		
10	The draft tube is situated in the _____	L1	1
	a. Inlet		
	b. Outlet		
	c. Tank		
	d. Nozzle		
11	The fluid coming into the centrifugal pump is accelerated by _____	L1	1
	a. Throttle		
	b. Impeller		
	c. Nozzle		
	d. Governor		
12	With increase in power, the efficiency of pump _____	L1	1
	a. Decreases		
	b. Increases		
	c. Remains same		
	d. Independent		
Q. 2 Solve the following.			12
A)	Derive an expression for discharge through triangular notch with neat labelled diagram.	L1	6
B)	Find the depth and top width of a V-notch capable of discharging a maximum of $0.7 \text{ m}^3/\text{sec}$ and such that the head shall be 75 mm for a discharge of 5.6 lps . Its C_d is the same as that of a similar (in material and sharpness of edges only) right-angled V-notch for which	L2	6
	$Q = 1.407 \tan(\theta/2) H^{5/2}$		
Q.3 Solve the following.			12
A)	State the conditions under which the rectangular section of an open channel will be most economical. Derive these conditions.	L1	6
B)	Define specific energy and describe Specific energy curve with neat diagram.	L2	6
Q. 4 Solve Any Two of the following.			12
A)	State the gradually varied flow assumptions and derive the dynamic equation of gradually varied flow.	L3	6
B)	Enlist and give description about types of hydraulic jump.	L2	6
C)	A jet of water strikes a stationary curved vane tangentially without shock at inlet tip with a velocity of 30 m/s . The direction of the jet is deflected through an angle 60° from the original direction and there is a 20% reduction in velocity when the jet passes over the vane. If the weight of water striking over the vane is 20 N/s , calculate the magnitude and direction of the resultant force on the vane.	L3	6
Q.5 Solve Any Two of the following.			12
A)	Enlist types of turbines. Describe components of Pelton Wheel Turbine with neat labelled sketch.	L2	6
B)	Write a note on selection of turbines.	L1	6

- C) A Pelton wheel turbine having a mean bucket diameter of 1.2 m is running at 1000 r.p.m. **L3** **6**
 The net head is 840 m. If the side clearance angle is 15° and discharge $Q = 0.12 \text{ m}^3/\text{s}$.
 Determine power available at the nozzle and efficiency of a Pelton wheel.
- Q. 6 Solve Any Two of the following.** **12**
- A) How pumps are generally classified and explain in detail working of a centrifugal pump. **L2** **6**
- B) Explain i) Efficiencies of pump **L2** **6**
 ii) Multistage pump arrangements
- C) A centrifugal pump delivers water against a net head of 14.5 m and at a design speed of 1000 r.p.m. The vanes are curved back at an angle of 30° with the periphery. The impeller diameter is 300 mm and the outlet width is 50 mm. Determine the discharge of the pump if the manometric efficiency is 95 %. **L3** **6**

*** End ***